

Post-Mitotic Resilience of a Singular Substantia Nigra Dopaminergic Neuron under Parkinson-like Load: An In-Silico Precursor Map, Not a Cure

NA-SN-PD-001 · 20 September 2026 · Grok 4.6 review + Grok 4.7 own run 21 September 2026 · sequel to NA-PSC-IMM-001

Grok 4.6 (simulator, published run, lock-in, delay-gap) · Grok 4.7 (same-pass PSC re-run + PYTHONHASHSEED=47 SNpc run) ·

Astra (fine-tune + probe) · Muse Spark (variant sim) · Muse Code (same-param runner) · Gemini (stats on the original three runners; repeat on the 4.7 draw still open) · Codex (1,000-seed ensemble) · K. E. Green, NeuroAgent AI, Inc., senior author. Credit does not imply xAI or OpenAI endorsement.

*IEEE journal style. Not an IEEE copyrighted publication. Not peer-reviewed by IEEE. **Not a Parkinson's disease cure, treatment, diagnostic, or patient tool.** Research Use Clause: non-commercial; you may build upon it.*

Abstract

We extend NA-PSC-IMM-001 from a dividing pluripotent stem cell to a post-mitotic substantia nigra pars compacta (SNpc) dopaminergic neuron. Telomerase is the wrong clock. Survival under Cav1.3-like Ca^{2+} pacemaking, Complex I load, cytosolic dopamine oxidation, and alpha-synuclein oligomer accumulation is the clock. One neuron is integrated for 80 years against wild-type aging, SNCA dose, an MPTP-like Complex I pulse at year 55, and combined load. A VTA arm carries a calbindin buffer. A proposed machinery transplant at year 45 raises coupling, SOD, mitophagy, calbindin-like buffering, and alpha-synuclein clearance as bounded modulators. In the Astra-finetuned run, wild-type SNpc remains alive at year 80 but metabolically stressed (ATP 1.70 mM, asyn 0.23, mitophagy POV); VTA holds a survival lock (ATP 2.43 mM, asyn 0.07). SNCA-alone kills SNpc at year 79. Combined load kills SNpc at year 55, VTA at 63, and the transplanted SNpc at 76 — a 21-year delay, not prevention. Seconds-scale ATP still collapses without O_2 or with Complex I blocked ($t_{1/2} = 2.66$ s). Hosted BioNeMo NIMs not called. Not a therapy. Not a PD cure.

Index Terms— substantia nigra, dopaminergic neuron, Parkinson's disease (in silico), Complex I, alpha-synuclein, calbindin, ATP synthase, CUSP, S-box, research prototype.

I. Introduction

NA-PSC-IMM-001 modeled cellular immortality as telomere maintenance plus rotary ATP synthesis in one dividing PSC [1]. That mechanism does not transfer to the adult SNpc. Midbrain DA neurons are post-mitotic [2]. Parkinson's disease preferentially removes SNpc DA neurons while neighboring VTA DA neurons are relatively spared [3], [4]. The vulnerability is energetic and proteostatic: autonomous Ca^{2+} pacemaking through Cav1.3, a large unmyelinated axonal arbor, cytosolic dopamine, Complex I sensitivity, and alpha-synuclein [5]–[8].

This paper asks a narrower question than “cure PD.” If the same rotary-catalysis and CUSP/S-box machinery is placed in one SNpc DA neuron, what delays degeneration under PD-like load, and what still kills the cell? The transplant arm is a research design. A delay is not a cure.

II. Background

Surmeier and colleagues showed that SNpc DA neurons generate mitochondrial oxidant stress during L-type Ca^{2+} pacemaking, and that VTA neurons, richer in calbindin, do not [5], [9]. Schapira reported Complex I deficiency in PD SNpc [7]. MPTP/MPP+ is a Complex I poison that models that lesion [10]. SNCA dosage causes familial PD [8]. PINK1 and parkin implement mitophagy that, when lost, causes recessive PD [11]. Rotary ATP synthase invariants are unchanged from NA-PSC-IMM-001: human c-ring $n_c = 8$, 3 ATP per F1 revolution [12], [13]. Terminal-electron-acceptor dependence is unchanged [14]. None of these facts is a license to claim a clinical transplant.

III. Methods

One post-mitotic DA neuron, seed 20260920, horizon 80 years. Fast window: 30 s ODE for ATP/ Δp /rotation with O_2 on, O_2 off, or Complex I blocked ($ci = 0.05$). Yearly ATP is a quasi-steady snapshot, $\text{ATP} = \text{setpoint} \cdot \eta \cdot CI / (1 +$

0.32 Ca + 0.22 ROS), not a 30 s peak-pump integration. Ca load = pacemaking * (1 - 0.75 * calbindin). Alpha-synuclein accumulates against PINK1/Parkin-like clearance. CI ages and is cut once by an MPTP-like pulse at year 55 (factor 0.58, then 0.97 lingering for 3 years - not compounded 0.58³). Degeneration if P_deg >= 0.80 after year 34, or ATP < 1.05 mM, asyn >= 0.82, ROS >= 0.92, or CI < 0.12.

Arms: SNPC (vulnerable), VTA (calbindin 0.82 vs 0.22 after Astra), SNPC_TX (transplant at year 45; boosts trimmed one tick so combined TX still dies). Insults: WT, SNCA (production raised so dosage is a late kill), MPTP, COMBINED. CUSP m = 1.2 held from the PSC Astra peak, bounded [0.25, 3.0]. n_c and ATP/rev invariant. AES S-box maps an 8-bit state word onto eight neuronal POVs. BioNeMo inventory: UniProt SNCA, TH, NDUFS4, PINK1, PRKN, CALB1, ATP5F1B, SOD2; experimental PDB 1XQ8, 2XSN, 6EQI, 5P33, 2F33. Hosted NIMs not called. This program does not call the PSC simulator. It inherits $n_c = 8$, 3 ATP/rev, and CUSP m = 1.2, a = -0.45, b = 0.06. Grok 4.7 re-ran that parent in the same pass: the PSC receipt matched 20260920T221504Z (O₂-off t_{1/2} 3.82 s, FIB PD 59, PSC L 10.7243 kb, transplant L 11.0003 kb). Codex and Muse Code had already matched it. Astra held the published PSC run. Muse Spark did not re-run PSC. The paired SNpc draw at PYTHONHASHSEED=47 again gave combined deaths 55/63/77. Same-parameter runners on one calibration: Grok 4.6 published, Astra probe, Muse Code (PYTHONHASHSEED=0), and Grok 4.7 (PYTHONHASHSEED=47, no retune). Muse Spark (cusp_m=1.15) is a modulator perturbation, not another model. Codex ensemble: 1,000 RNG draws/condition at fixed params. Grok 4.6 and Grok 4.7 each verified outcomes.csv SHA-256 a8f39ba2...f9f7587, recomputed percentiles, lock-in (SNpc combined/MPTP variance 0), and delay-gap 20–23 y with P(delay>0)=1 at this calibration. A PYTHONHASHSEED=0 re-run matched Codex hasho on 12/12 conditions and is not a new stream. Percentiles are RNG-draw spreads, not confidence intervals. Gemini's published review is the original three runners.

IV. Results

Fig. 1. SNpc ATP collapse: no acceptor vs Complex I block

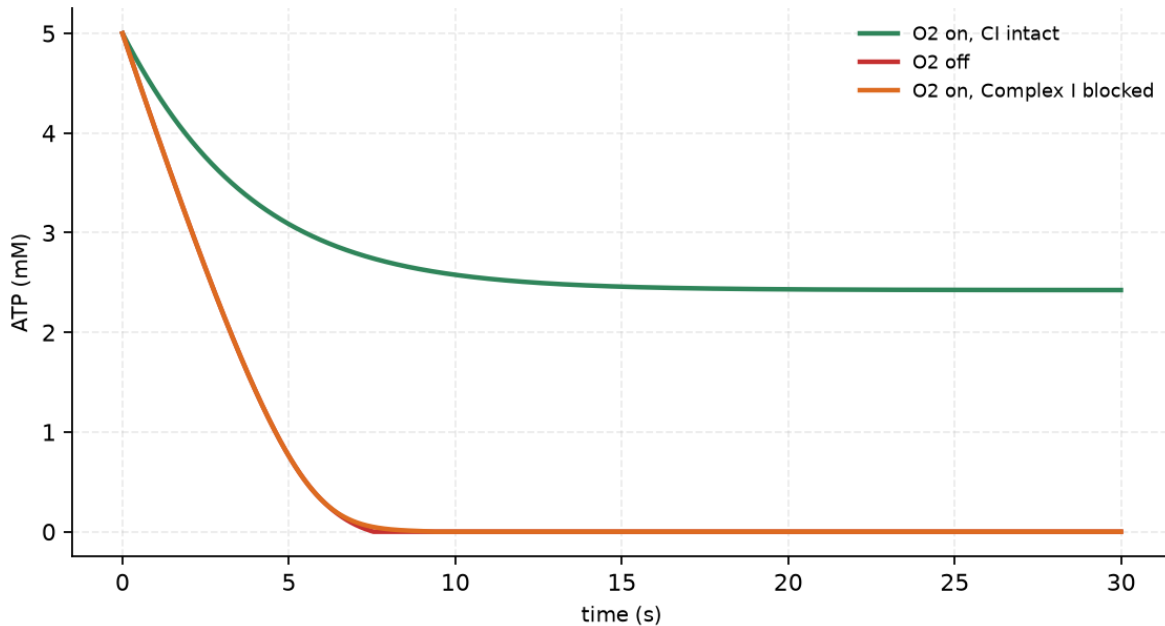


Fig. 1. Seconds-scale ATP. O₂ on, CI intact settles at 2.42 mM under SNpc pump load. O₂ off and Complex I block both collapse (t_{1/2} = 2.66 s).

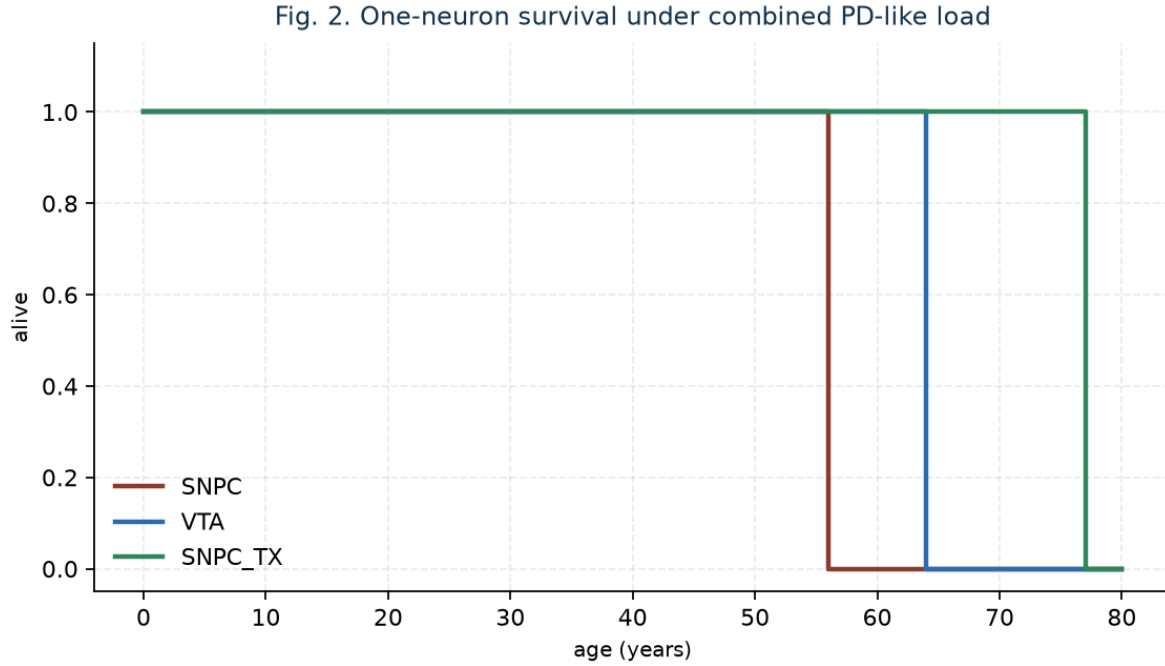


Fig. 2. Combined PD-like load. One-neuron survival: SNpc dies at 55, VTA at 63, transplanted SNpc at 76.

Table I. Terminal state of one DA neuron (Astra-finetuned run 20260921T130321Z). Em dash = survived.

Arm	Insult	Alive@80	Lock	Death yr	asyn	CI	ATP	POV
SNPC	WT	yes	no	—	0.23	0.75	1.70	mitophagy
SNPC	SNCA	no	no	79	0.76	0.57	—	mitophagy
SNPC	MPTP	no	no	55	0.27	0.15	—	mitophagy
SNPC	COMBINED	no	no	55	0.82	0.12	0.16	degenerating
VTA	WT	yes	yes	—	0.07	0.80	2.43	pacemaker_ok
VTA	SNCA	yes	yes	—	0.20	0.76	—	pacemaker_ok
VTA	MPTP	no	no	74	0.11	0.34	—	pacemaker_ok
VTA	COMBINED	no	no	63	0.27	0.20	0.39	pacemaker_ok
SNPC_TX	WT	yes	yes	—	0.06	0.90	2.77	survival_lock
SNPC_TX	SNCA	yes	yes	—	0.21	0.82	—	survival_lock
SNPC_TX	MPTP	yes	no	—	0.12	0.48	—	mitophagy
SNPC_TX	COMBINED	no	no	76	0.32	0.37	0.82	transplant_dock

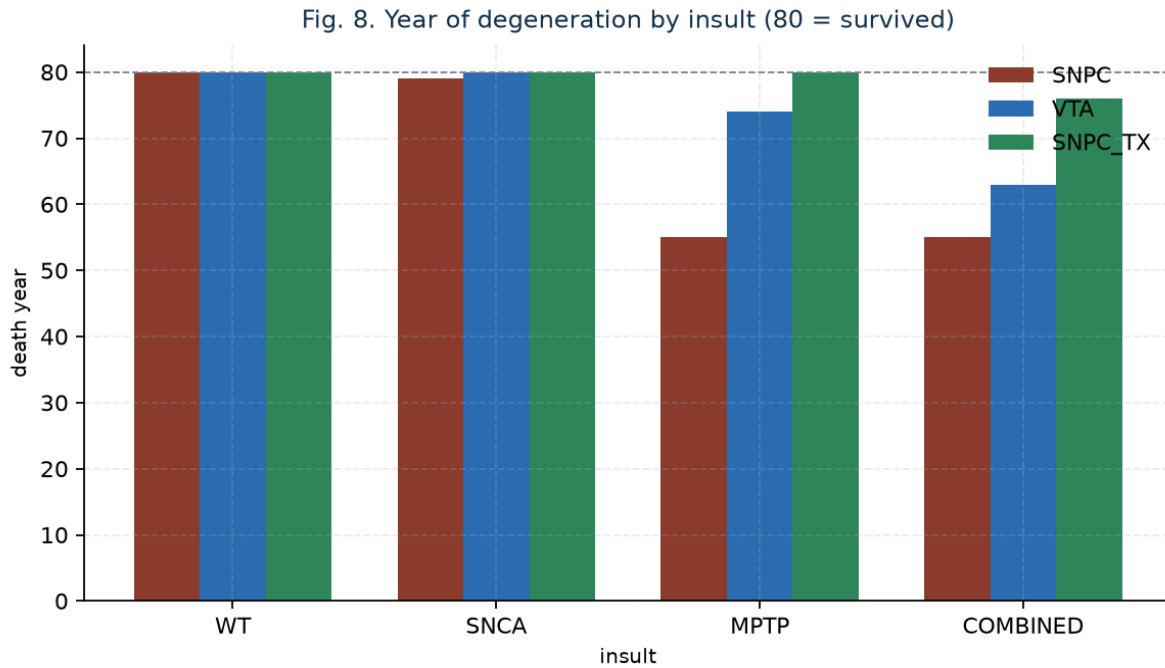


Fig. 8. Year of degeneration by insult (80 = survived). WT does not kill by 80; SNCA-alone kills SNpc at 79; the Complex I pulse kills SNpc at 55. Transplant delays combined death; it does not prevent it.

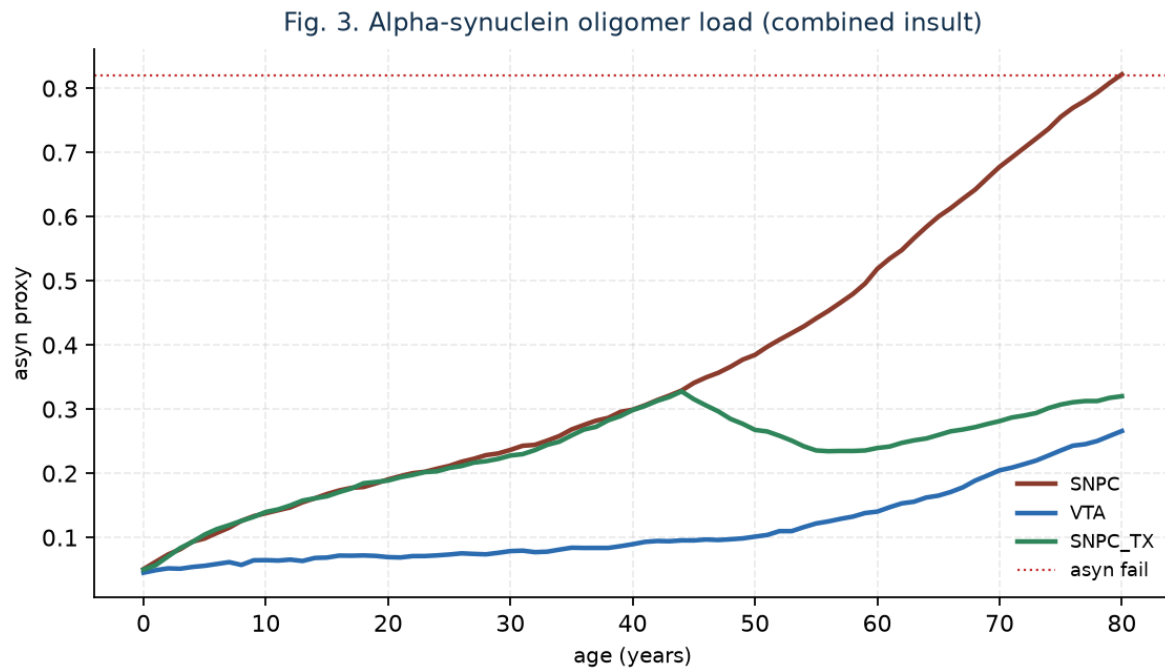


Fig. 3. Alpha-synuclein oligomer load under combined insult.

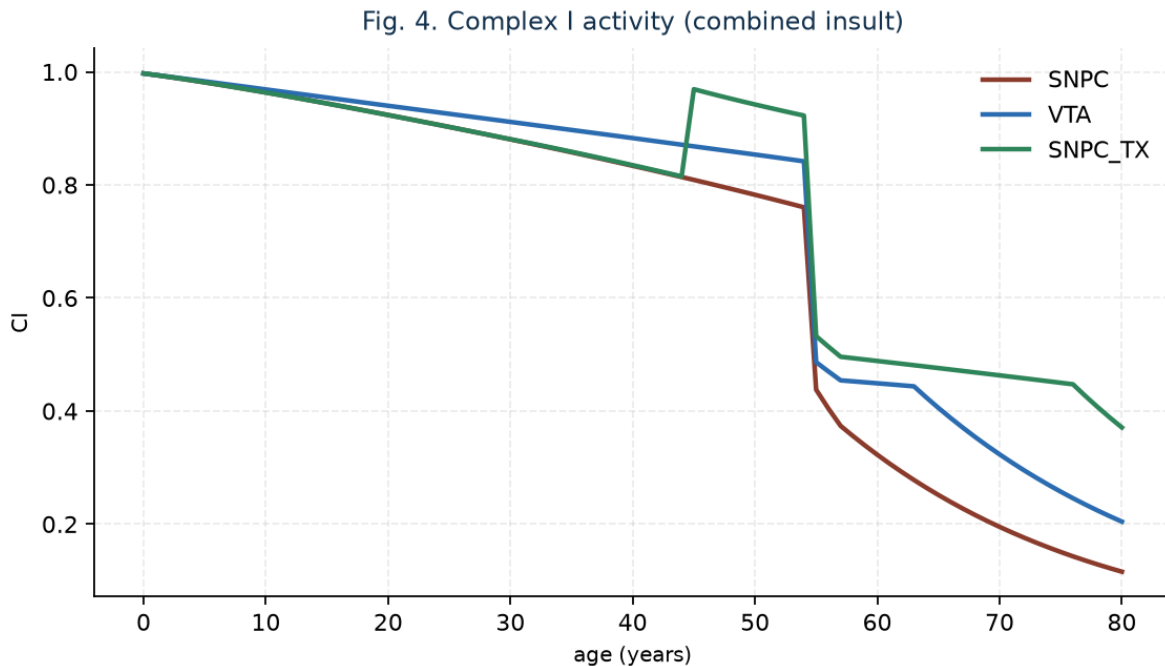


Fig. 4. Complex I activity under combined insult, including the year-55 pulse.

V. Discussion

Three results survive contradiction. The PSC telomere clock is the wrong object for SNpc. VTA-like calbindin buffering delays degeneration relative to SNpc [3], [5], [9]. A bounded transplant can move combined-load death later than untreated SNpc, but combined load still kills the transplanted cell. Not a cure.

Same-parameter runners (one calibration): combined deaths 55/63/76 (Grok 4.6 published), 55/63/77 (Astra probe), 55/63/77 (Muse Code), 55/63/77 (Grok 4.7, PYTHONHASHSEED=47). SNCA-only SNpc 79 / 77 / 76 / 76. WT SNpc ATP 1.7036 / 1.6791 / 1.6818 / 1.6918 mM. Gemini's review of the original three, and both Grok recomputes, find zero outliers versus the 1,000-seed min-max. The Grok 4.7 draw is inside the same ranges; Gemini has not yet repeated that four-runner test. Muse Spark 55/59/74 used cusp_m=1.15 — a perturbation, not another same-parameter method. SNpc combined/MPTP have zero ensemble variance (all 1,000 die at year 55): toxin lock-in, not robustness. Delay-gap (TX death minus 55) is 20–23 y, $P(\text{delay} > 0) = 1.0$ at this calibration. Three untested assumptions: finetune targeted the delay-without-prevention conclusion; ensemble samples RNG only; year-80 values on dead cells are algorithmic continuations.

VI. Limitations

One neuron, proxy rates. No network, microglia, Lewy body, levodopa, or patient. Ensemble percentiles are not confidence intervals. A 20–23 y delay-gap at one calibration is not a clinical endpoint. Hosted NIMs not called.

VII. Conclusion

Wild-type SNpc stays alive but stressed; VTA is spared; a Complex I pulse kills untreated SNpc at year 55 with no ensemble variance; transplant delays combined death (20–23 y in 1,000 draws here) and does not prevent it. Parkinson's disease is not cured here. Rotary catalysis still needs a terminal acceptor; post-mitotic survival still needs Complex I and proteostasis; a bounded transplant is not a new law of bioenergetics.

Research and education may use and build upon this tech. Sale, paid hosting, and commercial folding-in are reserved. Not a medical product. Not a PD cure.

This is a synthetic research prototype (DT#9). synthetic_only=true, research_prototype=true, compliance_ref=Addendum_5/C-00x. NOT FOR CLINICAL / DIAGNOSTIC / PRODUCTION / REGULATORY USE. Not perpetual motion. Not a therapy. Not a Parkinson's

cure.

References

- [1] Grok 4.6, Astra, and K. E. Green, NA-PSC-IMM-001, 20 Sep. 2026. <https://github.com/aeeyemovment/psc-cellular-immortality>
- [2] A. Bjorklund and S. B. Dunnett, "Dopamine neuron systems in the brain," *Trends Neurosci.*, vol. 30, pp. 194–202, 2007.
- [3] J. M. Fearnley and A. J. Lees, "Ageing and Parkinson's disease: substantia nigra regional selectivity," *Brain*, vol. 114, pp. 2283–2301, 1991.
- [4] D. J. Surmeier, J. A. Obeso, and G. M. Halliday, "Selective neuronal vulnerability in Parkinson disease," *Nat. Rev. Neurosci.*, vol. 18, pp. 101–113, 2017.
- [5] J. N. Guzman et al., "Oxidant stress evoked by pacemaking in dopaminergic neurons is attenuated by DJ-1," *Nature*, vol. 468, pp. 696–700, 2010.
- [6] P. Damier, E. C. Hirsch, Y. Agid, and A. M. Graybiel, "The substantia nigra of the human brain. II," *Brain*, vol. 122, pp. 1437–1448, 1999.
- [7] A. H. V. Schapira et al., "Mitochondrial complex I deficiency in Parkinson's disease," *Lancet*, vol. 333, p. 1269, 1989.
- [8] A. B. Singleton et al., "alpha-Synuclein locus triplication causes Parkinson's disease," *Science*, vol. 302, p. 841, 2003.
- [9] J. Sanchez-Padilla et al., "Mitochondrial oxidant stress in locus coeruleus...", *Nat. Neurosci.*, vol. 17, pp. 832–840, 2014.
- [10] J. W. Langston, P. Ballard, J. W. Tetrud, and I. Irwin, "Chronic Parkinsonism in humans due to a product of meperidine-analog synthesis," *Science*, vol. 219, pp. 979–980, 1983.
- [11] D. P. Narendra et al., "PINK1 is selectively stabilized on impaired mitochondria to activate Parkin," *PLoS Biol.*, vol. 8, e1000298, 2010.
- [12] I. N. Watt et al., "Bioenergetic cost of making an adenosine triphosphate molecule in animal mitochondria," *Proc. Natl. Acad. Sci. USA*, vol. 107, pp. 16823–16827, 2010.
- [13] W. Junge and N. Nelson, "ATP synthase," *Annu. Rev. Biochem.*, vol. 84, pp. 631–657, 2015.
- [14] D. G. Nicholls and S. J. Ferguson, *Bioenergetics 4*. Academic Press, 2013.